1. Description

1.1. Project

Project Name	UMH Controller V6
Board Name	custom
Generated with:	STM32CubeMX 6.11.1
Date	04/01/2026

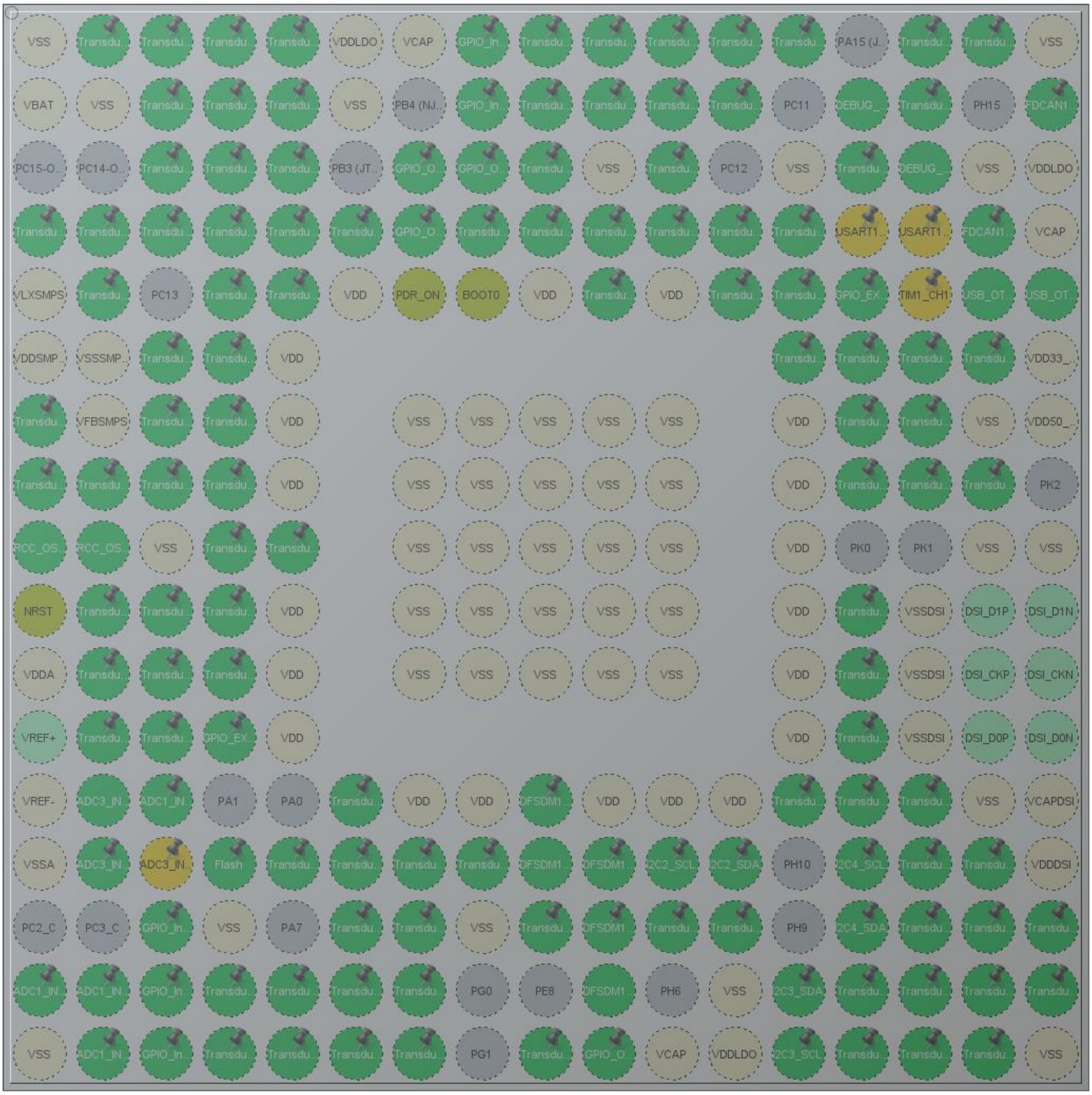
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H747/757
MCU name	STM32H757XIHx
MCU Package	TFBGA240
MCU Pin number	265

1.3. Core(s) information

Core(s)	ARM Cortex-M7 ARM Cortex-M4

2. Pinout Configuration



TFBGA240 +25 (Top view)

3. Pins Configuration

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A1	VSS	Power		
A2	PI6 *	I/O	GPIO_Output	Transducer
A3	PI5 *	I/O	GPIO_Output	Transducer
A4	PI4 *	I/O	GPIO_Output	Transducer
A5	PB5 *	I/O	GPIO_Output	Transducer
A6	VDDLDO	Power		
A7	VCAP	Power		
A8	PK5 *	I/O	GPIO_Input	
A9	PG10 *	I/O	GPIO_Output	Transducer
A10	PG9 *	I/O	GPIO_Output	Transducer
A11	PD5 *	I/O	GPIO_Output	Transducer
A12	PD4 *	I/O	GPIO_Output	Transducer
A13	PC10 *	I/O	GPIO_Output	Transducer
A15	PI1 *	I/O	GPIO_Output	Transducer
A16	PI0 *	I/O	GPIO_Output	Transducer
A17	VSS	Power		
B1	VBAT	Power		
B2	VSS	Power		
B3	PI7 *	I/O	GPIO_Output	Transducer
B4	PE1 *	I/O	GPIO_Output	Transducer
B5	PB6 *	I/O	GPIO_Output	Transducer
B6	VSS	Power		
B8	PK4 *	I/O	GPIO_Input	
B9	PG11 *	I/O	GPIO_Output	Transducer
B10	PJ15 *	I/O	GPIO_Output	Transducer
B11	PD6 *	I/O	GPIO_Output	Transducer
B12	PD3 *	I/O	GPIO_Output	Transducer
B14	PA14 (JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
B15	PI2 *	I/O	GPIO_Output	Transducer
B17	PH14	I/O	FDCAN1_RX	
C3	PE2 *	I/O	GPIO_Output	Transducer
C4	PE0 *	I/O	GPIO_Output	Transducer
C5	PB7 *	I/O	GPIO_Output	Transducer
C7	PK6 *	I/O	GPIO_Output	
C8	PK3 *	I/O	GPIO_Output	
C9	PG12 *	I/O	GPIO_Output	Transducer

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
C10	VSS	Power		
C11	PD7 *	I/O	GPIO_Output	Transducer
C13	VSS	Power		
C14	PI3 *	I/O	GPIO_Output	Transducer
C15	PA13 (JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
C16	VSS	Power		
C17	VDDLDO	Power		
D1	PE5 *	I/O	GPIO_Output	Transducer
D2	PE4 *	I/O	GPIO_Output	Transducer
D3	PE3 *	I/O	GPIO_Output	Transducer
D4	PB9 *	I/O	GPIO_Output	Transducer
D5	PB8 *	I/O	GPIO_Output	Transducer
D6	PG15 *	I/O	GPIO_Output	Transducer
D7	PK7 *	I/O	GPIO_Output	
D8	PG14 *	I/O	GPIO_Output	Transducer
D9	PG13 *	I/O	GPIO_Output	Transducer
D10	PJ14 *	I/O	GPIO_Output	Transducer
D11	PJ12 *	I/O	GPIO_Output	Transducer
D12	PD2 *	I/O	GPIO_Output	Transducer
D13	PD0 *	I/O	GPIO_Output	Transducer
D14	PA10 **	I/O	USART1_RX	
D15	PA9 **	I/O	USART1_TX	
D16	PH13	I/O	FDCAN1_TX	
D17	VCAP	Power		
E1	VLXSMPS	Power		
E2	PI9 *	I/O	GPIO_Output	Transducer
E4	PI8 *	I/O	GPIO_Output	Transducer
E5	PE6 *	I/O	GPIO_Output	Transducer
E6	VDD	Power		
E7	PDR_ON	Reset		
E8	BOOT0	Boot		
E9	VDD	Power		
E10	PJ13 *	I/O	GPIO_Output	Transducer
E11	VDD	Power		
E12	PD1 *	I/O	GPIO_Output	Transducer
E13	PC8 *	I/O	GPIO_Output	Transducer
E14	PC9	I/O	GPIO_EXTI9	
E15	PA8 **	I/O	TIM1_CH1	
E16	PA12	I/O	USB_OTG_FS_DP	

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
E17	PA11	I/O	USB_OTG_FS_DM	
F1	VDDSMPS	Power		
F2	VSSSMPS	Power		
F3	PI10 *	I/O	GPIO_Output	Transducer
F4	PI11 *	I/O	GPIO_Output	Transducer
F5	VDD	Power		
F13	PC7 *	I/O	GPIO_Output	Transducer
F14	PC6 *	I/O	GPIO_Output	Transducer
F15	PG8 *	I/O	GPIO_Output	Transducer
F16	PG7 *	I/O	GPIO_Output	Transducer
F17	VDD33_USB	Power		
G1	PF2 *	I/O	GPIO_Output	Transducer
G2	VFBSMPS	Power		
G3	PF1 *	I/O	GPIO_Output	Transducer
G4	PF0 *	I/O	GPIO_Output	Transducer
G5	VDD	Power		
G7	VSS	Power		
G8	VSS	Power		
G9	VSS	Power		
G10	VSS	Power		
G11	VSS	Power		
G13	VDD	Power		
G14	PG5 *	I/O	GPIO_Output	Transducer
G15	PG6 *	I/O	GPIO_Output	Transducer
G16	VSS	Power		
G17	VDD50_USB	Power		
H1	PI12 *	I/O	GPIO_Output	Transducer
H2	PI13 *	I/O	GPIO_Output	Transducer
H3	PI14 *	I/O	GPIO_Output	Transducer
H4	PF3 *	I/O	GPIO_Output	Transducer
H5	VDD	Power		
H7	VSS	Power		
H8	VSS	Power		
H9	VSS	Power		
H10	VSS	Power		
H11	VSS	Power		
H13	VDD	Power		
H14	PG4 *	I/O	GPIO_Output	Transducer
H15	PG3 *	I/O	GPIO_Output	Transducer

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
H16	PG2 *	I/O	GPIO_Output	Transducer
J1	PH1-OSC_OUT (PH1)	I/O	RCC_OSC_OUT	
J2	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
J3	VSS	Power		
J4	PF5 *	I/O	GPIO_Output	Transducer
J5	PF4 *	I/O	GPIO_Output	Transducer
J7	VSS	Power		
J8	VSS	Power		
J9	VSS	Power		
J10	VSS	Power		
J11	VSS	Power		
J13	VDD	Power		
J16	VSS	Power		
J17	VSS	Power		
K1	NRST	Reset		
K2	PF6 *	I/O	GPIO_Output	Transducer
K3	PF7 *	I/O	GPIO_Output	Transducer
K4	PF8 *	I/O	GPIO_Output	Transducer
K5	VDD	Power		
K7	VSS	Power		
K8	VSS	Power		
K9	VSS	Power		
K10	VSS	Power		
K11	VSS	Power		
K13	VDD	Power		
K14	PJ11 *	I/O	GPIO_Output	Transducer
K15	VSSDSI	Power		
L1	VDDA	Power		
L2	PC0 *	I/O	GPIO_Output	Transducer
L3	PF10 *	I/O	GPIO_Output	Transducer
L4	PF9 *	I/O	GPIO_Output	Transducer
L5	VDD	Power		
L7	VSS	Power		
L8	VSS	Power		
L9	VSS	Power		
L10	VSS	Power		
L11	VSS	Power		
L13	VDD	Power		
L14	PJ10 *	I/O	GPIO_Output	Transducer

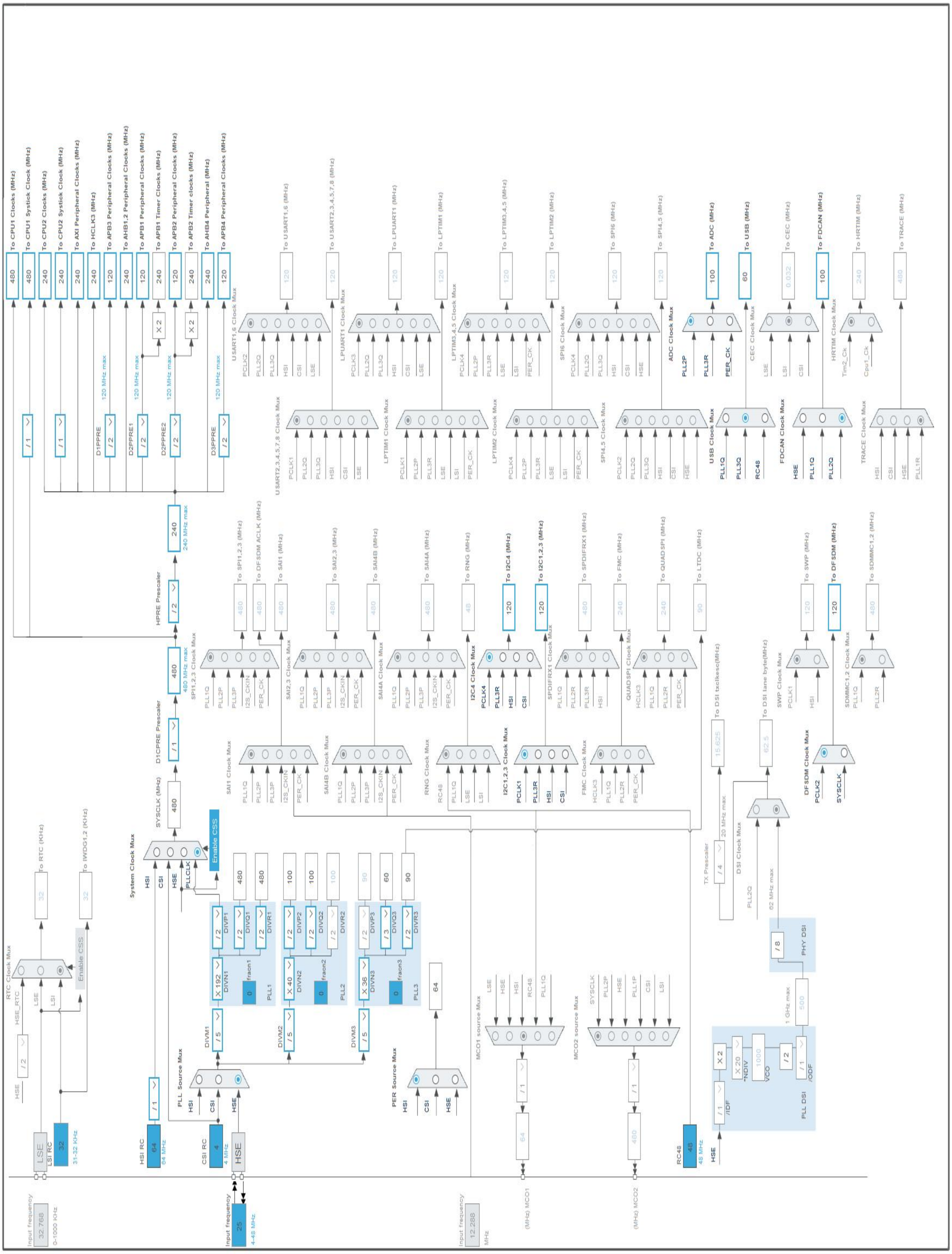
Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
L15	VSSDSI	Power		
M2	PC1 *	I/O	GPIO_Output	Transducer
M3	PC2 *	I/O	GPIO_Output	Transducer
M4	PC3	I/O	GPIO_EXTI3	
M5	VDD	Power		
M13	VDD	Power		
M14	PJ9 *	I/O	GPIO_Output	Transducer
M15	VSSDSI	Power		
N1	VREF-	Power		
N2	PH2	I/O	ADC3_INP13	
N3	PA2	I/O	ADC1_INP14	
N6	PJ0 *	I/O	GPIO_Output	Transducer
N7	VDD	Power		
N8	VDD	Power		
N9	PE10	I/O	DFSDM1_DATIN4	
N10	VDD	Power		
N11	VDD	Power		
N12	VDD	Power		
N13	PJ8 *	I/O	GPIO_Output	Transducer
N14	PJ7 *	I/O	GPIO_Output	Transducer
N15	PJ6 *	I/O	GPIO_Output	Transducer
N16	VSS	Power		
N17	VCAPDSI	Power		
P1	VSSA	Power		
P2	PH3	I/O	ADC3_INP14	
P3	PH4 **	I/O	ADC3_INP15	
P4	PH5 *	I/O	GPIO_Output	Flash
P5	PI15 *	I/O	GPIO_Output	Transducer
P6	PJ1 *	I/O	GPIO_Output	Transducer
P7	PF13 *	I/O	GPIO_Output	Transducer
P8	PF14 *	I/O	GPIO_Output	Transducer
P9	PE9	I/O	DFSDM1_CKOUT	
P10	PE11	I/O	DFSDM1_CKIN4	
P11	PB10	I/O	I2C2_SCL	
P12	PB11	I/O	I2C2_SDA	
P14	PH11	I/O	I2C4_SCL	
P15	PD15 *	I/O	GPIO_Output	Transducer
P16	PD14 *	I/O	GPIO_Output	Transducer
P17	VDDDSI	Power		

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
R3	PA6 *	I/O	GPIO_Input	
R4	VSS	Power		
R6	PB2 *	I/O	GPIO_Output	Transducer
R7	PF12 *	I/O	GPIO_Output	Transducer
R8	VSS	Power		
R9	PF15 *	I/O	GPIO_Output	Transducer
R10	PE12	I/O	DFSDM1_DATIN5	
R11	PE15 *	I/O	GPIO_Output	Transducer
R12	PJ5 *	I/O	GPIO_Output	Transducer
R14	PH12	I/O	I2C4_SDA	
R15	PD11 *	I/O	GPIO_Output	Transducer
R16	PD12 *	I/O	GPIO_Output	Transducer
R17	PD13 *	I/O	GPIO_Output	Transducer
T1	PA0_C	I/O	ADC1_INP0	
T2	PA1_C	I/O	ADC1_INP1	
T3	PA5 *	I/O	GPIO_Input	
T4	PC4 *	I/O	GPIO_Output	Transducer
T5	PB1 *	I/O	GPIO_Output	Transducer
T6	PJ2 *	I/O	GPIO_Output	Transducer
T7	PF11 *	I/O	GPIO_Output	Transducer
T10	PE13	I/O	DFSDM1_CKIN5	
T12	VSS	Power		
T13	PH8	I/O	I2C3_SDA	
T14	PB12 *	I/O	GPIO_Output	Transducer
T15	PB15 *	I/O	GPIO_Output	Transducer
T16	PD10 *	I/O	GPIO_Output	Transducer
T17	PD9 *	I/O	GPIO_Output	Transducer
U1	VSS	Power		
U2	PA3	I/O	ADC1_INP15	
U3	PA4 *	I/O	GPIO_Input	
U4	PC5 *	I/O	GPIO_Output	Transducer
U5	PB0 *	I/O	GPIO_Output	Transducer
U6	PJ3 *	I/O	GPIO_Output	Transducer
U7	PJ4 *	I/O	GPIO_Output	Transducer
U9	PE7 *	I/O	GPIO_Output	Transducer
U10	PE14 *	I/O	GPIO_Output	
U11	VCAP	Power		
U12	VDDLDO	Power		
U13	PH7	I/O	I2C3_SCL	

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
U14	PB13 *	I/O	GPIO_Output	Transducer
U15	PB14 *	I/O	GPIO_Output	Transducer
U16	PD8 *	I/O	GPIO_Output	Transducer
U17	VSS	Power		

* The pin is affected with an I/O function
** The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration



5. Software Project

5.1. Project Settings

Name	Value
Project Name	UMH Controller V6
Project Folder	D:\Data\OneDrive\Projects\UMH\Software\UMH Controller
Toolchain / IDE	EWARM V8.50
Firmware Package Name and Version	STM32Cube FW_H7 V1.11.2
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	M4-0x200
Minimum Stack Size	M4-0x400

5.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

5.3. Advanced Settings - Generated Function Calls ARM Cortex-M7

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DFSDM1_Init	DFSDM1

5.4. Advanced Settings - Generated Function Calls ARM Cortex-M4

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	MX_USB_DEVICE_Init	USB_DEVICE_M4
3	MX_ADC1_Init	ADC1

Rank	Function Name	Peripheral Instance Name
4	MX_ADC3_Init	ADC3
5	MX_FDCAN1_Init	FDCAN1
6	MX_I2C2_Init	I2C2
7	MX_I2C3_Init	I2C3
8	MX_I2C4_Init	I2C4
9	MX_FREERTOS_Init	FREERTOS_M4

1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H747/757
MCU	STM32H757XIHx
Datasheet	DS12931_Rev1

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(DD36000)
Capacity	36000.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	450.0 mA
Max Pulse Current	1000.0 mA
Cells in series	1
Cells in parallel	1

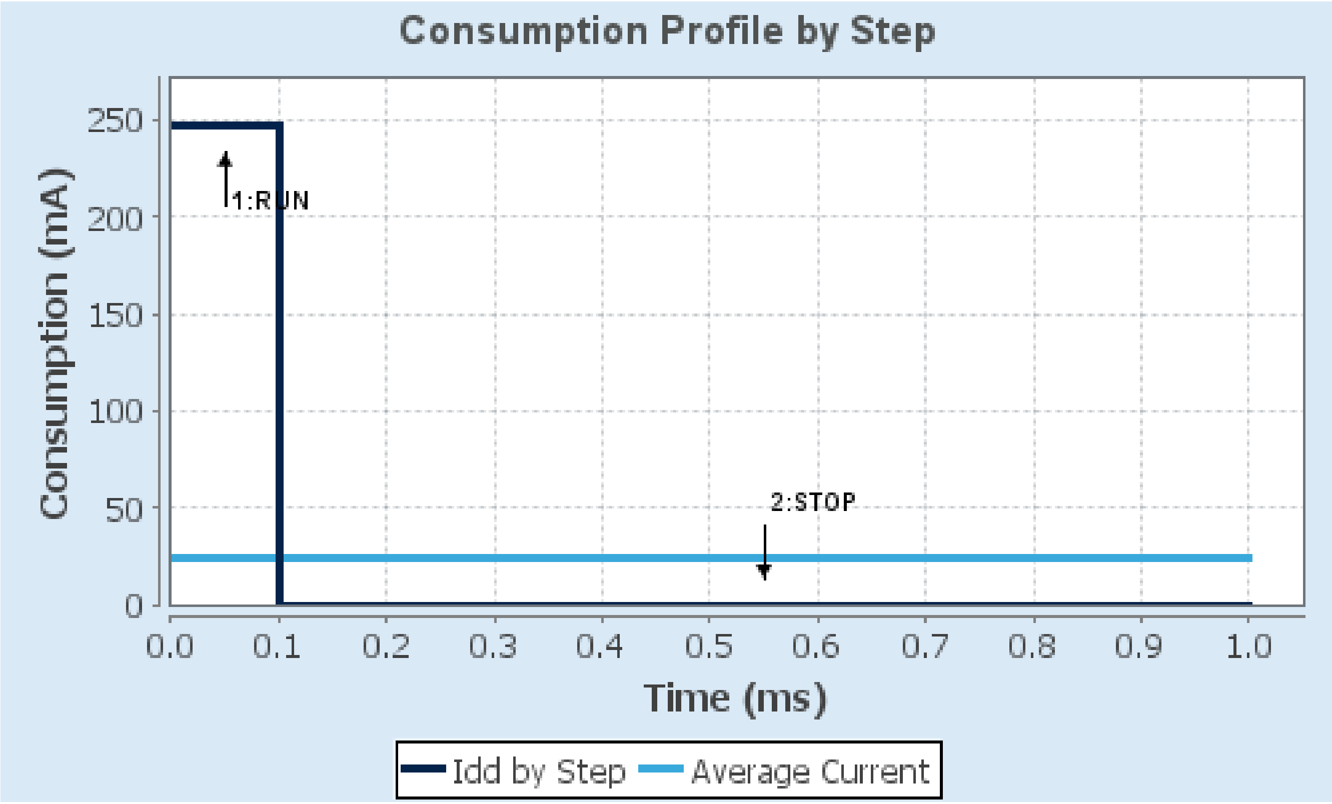
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN/CRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	CM7: ITCM/Cache / CM4: FLASH_B/ART	CM7: NA / CM4: NA
CM7 Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL ALL_IPs_ON	LSE Flash-ON
CM4 Frequency	240 MHz	0 Hz
Clock Source Frequency	25 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	247 mA	145 µA
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	24.83 mA
Battery Life	1 month, 29 days, 21 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
250
200
150
100
50
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Peripherals and Middlewares Configuration

2.1. ADC1

mode: IN0

IN1: IN1 Single-ended

mode: IN14

mode: IN15

2.1.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

ADCs_Common_Settings:

Mode	Independent mode

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1
Resolution	ADC 16-bit resolution
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Left Bit Shift	No bit shift
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Rank	1
Channel	**Channel 1 ***
Sampling Time	1.5 Cycles
Offset Number	No offset
Offset Signed Saturation	Disable

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

2.2. ADC3
IN13: IN13 Single-ended
IN14: IN14 Single-ended
2.2.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D3

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1
Resolution	ADC 16-bit resolution
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Left Bit Shift	No bit shift
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Rank	1
Channel	**Channel 14 ***
Sampling Time	1.5 Cycles
Offset Number	No offset
Offset Signed Saturation	Disable

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

2.3. CORTEX_M7

2.3.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Speculation default mode Settings:

Speculation default mode	**Enabled ***

Cortex Interface Settings:

CPU ICache	Disabled
CPU DCache	Disabled

Cortex Memory Protection Unit Control Settings:

MPU Control Mode	Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers

Cortex Memory Protection Unit Region 0 Settings:

MPU Region	Enabled
MPU Region Base Address	**0x0 ***
MPU Region Size	4GB
MPU SubRegion Disable	**0x87 ***
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS NOT PERMITTED
MPU Instruction Access	DISABLE
MPU Shareability Permission	ENABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 2 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 3 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 4 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 5 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 6 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region	Disabled

2.4. DEBUG

Debug: Serial Wire

2.4.1. Core(s) Settings:

Context(s):	Cortex-M7 Cortex-M4
Initialized Context:	Cortex-M7
Power Domain:	

2.5. DFSDM1

mode: PDM/SPI input from ch4 and external clock

mode: PDM/SPI input from ch5 and external clock

mode: CKOUT

2.5.1. Filter 0:

Core(s) Settings:	
Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2
regular channel selection:	
regular channel selection	**Channel 4 ***
Continuous Mode	Continuous Mode
Trigger to start regular conversion	Software trigger
Fast Mode	Disable
Dma Mode	Disable
injected channel selection:	
Channel0 as injected channel	Disable
Channel1 as injected channel	Disable
Channel2 as injected channel	Disable
Channel3 as injected channel	Disable
Channel4 as injected channel	Disable
Channel5 as injected channel	Disable
Channel6 as injected channel	Disable
Channel7 as injected channel	Disable
Filter parameters:	
Sinc Order	FastSinc filter type
Fosr	1
Iosr	1

2.5.2. Filter 1:

Core(s) Settings:	
Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	

	D2

regular channel selection:

regular channel selection	- None -

injected channel selection:

Channel0 as injected channel	Disable
Channel1 as injected channel	Disable
Channel2 as injected channel	Disable
Channel3 as injected channel	Disable
Channel4 as injected channel	Disable
Channel5 as injected channel	Disable
Channel6 as injected channel	Disable
Channel7 as injected channel	Disable

2.5.3. Filter 2:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

regular channel selection:

regular channel selection	- None -

injected channel selection:

Channel0 as injected channel	Disable
Channel1 as injected channel	Disable
Channel2 as injected channel	Disable
Channel3 as injected channel	Disable
Channel4 as injected channel	Disable
Channel5 as injected channel	Disable
Channel6 as injected channel	Disable
Channel7 as injected channel	Disable

2.5.4. Filter 3:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

regular channel selection:

regular channel selection	- None -

injected channel selection:

Channel0 as injected channel	Disable
Channel1 as injected channel	Disable
Channel2 as injected channel	Disable
Channel3 as injected channel	Disable
Channel4 as injected channel	Disable
Channel5 as injected channel	Disable
Channel6 as injected channel	Disable
Channel7 as injected channel	Disable

2.5.5. Output Clock:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Output Clock parameters:

Selection	Source for ouput clock is system clock
Divider	2

2.5.6. Channel 4:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Channel 4 parameters:

Type	SPI with rising edge
Offset	0
Right Bit Shift	**0x00 ***

Analog watchdog parameters:

Filter Order	FastSinc filter type
Oversampling	1

2.5.7. Channel 5:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Analog watchdog parameters:

Filter Order	FastSinc filter type
Oversampling	1

Channel 5 parameters:

Type	SPI with rising edge
Offset	0
Right Bit Shift	**0x00 ***

2.6. FDCAN1
mode: Activated
2.6.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Basic Parameters:

Frame Format	Classic mode
Mode	Normal mode
Auto Retransmission	Disable
Transmit Pause	Disable
Protocol Exception	Disable
Nominal Sync Jump Width	1
Data Prescaler	1
Data Sync Jump Width	1
Data Time Seg1	1
Data Time Seg2	1
Message Ram Offset	0
Std Filters Nbr	0
Ext Filters Nbr	0
Rx Fifo0 Elmts Nbr	0
Rx Fifo0 Elmt Size	8 bytes data field

Rx Fifo1 Elmts Nbr	0
Rx Fifo1 Elmt Size	8 bytes data field
Rx Buffers Nbr	0
Rx Buffer Size	8 bytes data field
Tx Events Nbr	0
Tx Buffers Nbr	0
Tx Fifo Queue Elmts Nbr	0
Tx Fifo Queue Mode	FIFO mode
Tx Elmt Size	8 bytes data field

Clock Calibration Unit:

Clock Calibration	Disable

Bit Timings Parameters:

Nominal Prescaler	16
Nominal Time Quantum	**160.0 ***
Nominal Time Seg1	2
Nominal Time Seg2	2
Nominal Time for one Bit	**800 ***
Nominal Baud Rate	**1250000 ***

2.7. I2C2

I2C: I2C

2.7.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x307075B1 ***

Slave Features:

Clock No Stretch Mode	Disabled

General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

2.8. I2C3

I2C: I2C

2.8.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x307075B1 ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

2.9. I2C4

I2C: I2C

2.9.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4

Power Domain:	D3

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x307075B1 ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

2.10. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

2.10.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7 Cortex-M4
Initialized Context:	Cortex-M7
Power Domain:	D3

Power Parameters:

SupplySource	PWR_LDO_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000
CSI Calibration Value	32
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	4 WS (5 CPU cycle)

Product revision	rev.V

PLL range Parameters:

PLL1 clock Input range	Between 4 and 8 MHz
PLL2 input frequency range	Between 4 and 8 MHz
PLL3 input frequency range	Between 4 and 8 MHz
PLL1 clock Output range	Wide VCO range
PLL2 clock Output range	Wide VCO range
PLL3 clock Output range	MEDIUM VCO range

2.11. SYS_M4
Timebase Source: SysTick
2.11.1. Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	

2.12. SYS
Timebase Source: SysTick
2.12.1. Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	

2.13. USB_OTG_FS
Mode: Device_Only
2.13.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Speed	Full Speed 12MBit/s
Enable internal IP DMA	Disabled
Low power	Disabled
Battery charging	Disabled
Link Power Management	Disabled
Use dedicated end point 1 interrupt	Disabled
VBUS sensing	Disabled
Signal start of frame	Disabled

2.14. FREERTOS_M4
Interface: CMSIS_V1
2.14.1. Config parameters:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

API:

FreeRTOS API	CMSIS v1

Versions:

FreeRTOS version	10.3.1
CMSIS-RTOS version	1.02

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Disabled

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemD2Clock
TICK_RATE_HZ	1000
MAX_PRIORITIES	7
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Disabled
USE_COUNTING_SEMAPHORES	Disabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled

ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Enabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled
Memory management settings:	
Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	15360
Memory Management scheme	heap_4
Hook function related definitions:	
USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled
Run time and task stats gathering related definitions:	
GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Disabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled
Co-routine related definitions:	
USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2
Software timer definitions:	
USE_TIMERS	Disabled
Interrupt nesting behaviour configuration:	
LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5
Added with 10.2.1 support:	
MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

2.14.2. Include parameters:

Core(s) Settings:	
Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2
Include definitions:	
vTaskPrioritySet	Enabled

uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Disabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Disabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Disabled
xTaskGetCurrentTaskHandle	Disabled
eTaskGetState	Disabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Disabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

2.14.3. Advanced settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled

Project settings (see parameter description first):

Use FW pack heap file	Enabled

2.15. USB_DEVICE_M4

Class For FS IP: Communication Device Class (Virtual Port Com)

2.15.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	

	Cortex-M4
Power Domain:	D2

Basic Parameters:

USBD_MAX_NUM_INTERFACES (Maximum number of supported interfaces)	1
USBD_MAX_NUM_CONFIGURATION (Maximum number of supported configuration)	1
USBD_MAX_STR_DESC_SIZ (Maximum size for the string descriptors)	512
USBD_SELF_POWERED (Enabled self power)	Enabled
USBD_DEBUG_LEVEL (USBD Debug Level)	0: No debug message
USBD_LPM_ENABLED (Link Power Management)	1: Link Power Management supported

Class Parameters:

USB CDC Rx Buffer Size	2048
USB CDC Tx Buffer Size	2048

2.15.2. Device Descriptor:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Device Descriptor:

VID (Vendor IDentifier)	1155
LANGID_STRING (Language Identifier)	English(United States)
MANUFACTURER_STRING (Manufacturer Identifier)	STMicroelectronics

Device Descriptor FS:

PID (Product IDentifier)	22336
PRODUCT_STRING (Product Identifier)	STM32 Virtual ComPort
CONFIGURATION_STRING (Configuration Identifier)	CDC Config
INTERFACE_STRING (Interface Identifier)	CDC Interface
DEVICE_SERIAL0 (Device Serial Number 0)	**0x11223344 ***
DEVICE_SERIAL1 (Device Serial Number 1)	**0x55667788 ***
DEVICE_SERIAL2 (Device Serial Number 2)	**0x12345678 ***

*** User modified value**

3. System Configuration

3.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
ADC1	PA2	ADC1_INP14	Analog mode	No pull-up and no pull-down	n/a		Cortex-M4	D2
	PA0_C	ADC1_INP0	Analog mode	No pull-up and no pull-down	n/a		Cortex-M4	D2
	PA1_C	ADC1_INP1	Analog mode	No pull-up and no pull-down	n/a		Cortex-M4	D2
	PA3	ADC1_INP15	Analog mode	No pull-up and no pull-down	n/a		Cortex-M4	D2
ADC3	PH2	ADC3_INP13	Analog mode	No pull-up and no pull-down	n/a		Cortex-M4	D3
	PH3	ADC3_INP14	Analog mode	No pull-up and no pull-down	n/a		Cortex-M4	D3
DEBUG	PA14 (JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PA13 (JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
DFSDM1	PE10	DFSDM1_DATIN4	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PE9	DFSDM1_CKOUT	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PE11	DFSDM1_CKIN4	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PE12	DFSDM1_DATIN5	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PE13	DFSDM1_CKIN5	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
FDCAN1	PH14	FDCAN1_RX	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***		Cortex-M4	D2
	PH13	FDCAN1_TX	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***		Cortex-M4	D2
I2C2	PB10	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D2
	PB11	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D2
I2C3	PH8	I2C3_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D2
	PH7	I2C3_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D2

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
I2C4	PH11	I2C4_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D3
	PH12	I2C4_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D3
RCC	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a		Cortex-M7* Cortex-M4	D3
	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a		Cortex-M7* Cortex-M4	D3
USB_OTG_FS	PA12	USB_OTG_FS_DP	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PA11	USB_OTG_FS_DM	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
Single Mapped Signals	PA10	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***			
	PA9	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***			
	PA8	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low			
	PH4	ADC3_INP15	Analog mode	No pull-up and no pull-down	n/a			
GPIO	PI6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PK5	GPIO_Input	Input mode	No pull-up and no pull-down	**n/a**		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PC10	GPIO_Output	Output Push Pull	No pull-up and no pull-		Transducer	Cortex-M7*	Cortex-M7*

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
				down	**Very High ***		Cortex-M4	Cortex-M4
	PI1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PK4	GPIO_Input	Input mode	No pull-up and no pull-down	**n/a**		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PK6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PK3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
	PI3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PK7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very**	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
					High *			
	PC8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PC9	GPIO_EXTI9	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	**n/a**		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PC7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PC6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very**	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
					High *			
	PG3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PC0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PC1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PC2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PC3	GPIO_EXTI3	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	**n/a**		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very**	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
					High *			
	PJ7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PH5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Flash	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PI15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PA6	GPIO_Input	Input mode	**Pull-up ***	**n/a**		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
	PA5	GPIO_Input	Input mode	**Pull-up ***	**n/a**		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PC4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PA4	GPIO_Input	Input mode	**Pull-up ***	**n/a**		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PC5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	Transducer	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4

* Initialized context

3.2. DMA configuration

nothing configured in DMA service

3.3. BDMA configuration

nothing configured in DMA service

3.4. MDMA configuration

nothing configured in DMA service

3.5. NVIC configuration

3.5.1. NVIC1

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
CM4 send event interrupt for CM7	unused		
FPU global interrupt	unused		
DFSDM1 filter0 global interrupt	unused		
DFSDM1 filter1 global interrupt	unused		
DFSDM1 filter2 global interrupt	unused		
DFSDM1 filter3 global interrupt	unused		
HSEM1 global interrupt	unused		
RAM ECC diagnostic global interrupt	unused		
Hold core interrupt	unused		

3.5.2. NVIC1 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

3.5.3. NVIC2

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
USB On The Go FS global interrupt	true	5	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
EXTI line3 interrupt	unused		
ADC1 and ADC2 global interrupts	unused		
FDCAN1 interrupt 0	unused		
FDCAN1 interrupt 1	unused		
EXTI line[9:5] interrupts	unused		
I2C2 event interrupt	unused		
I2C2 error interrupt	unused		
FDCAN calibration unit interrupt	unused		
CM7 send event interrupt for CM4	unused		
I2C3 event interrupt	unused		
I2C3 error interrupt	unused		
FPU global interrupt	unused		
I2C4 event interrupt	unused		
I2C4 error interrupt	unused		
USB On The Go FS End Point 1 Out global interrupt	unused		
USB On The Go FS End Point 1 In global interrupt	unused		
HSEM2 global interrupt	unused		
ADC3 global interrupt	unused		
RAM ECC diagnostic global interrupt	unused		
Hold core interrupt	unused		

3.5.4. NVIC2 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false

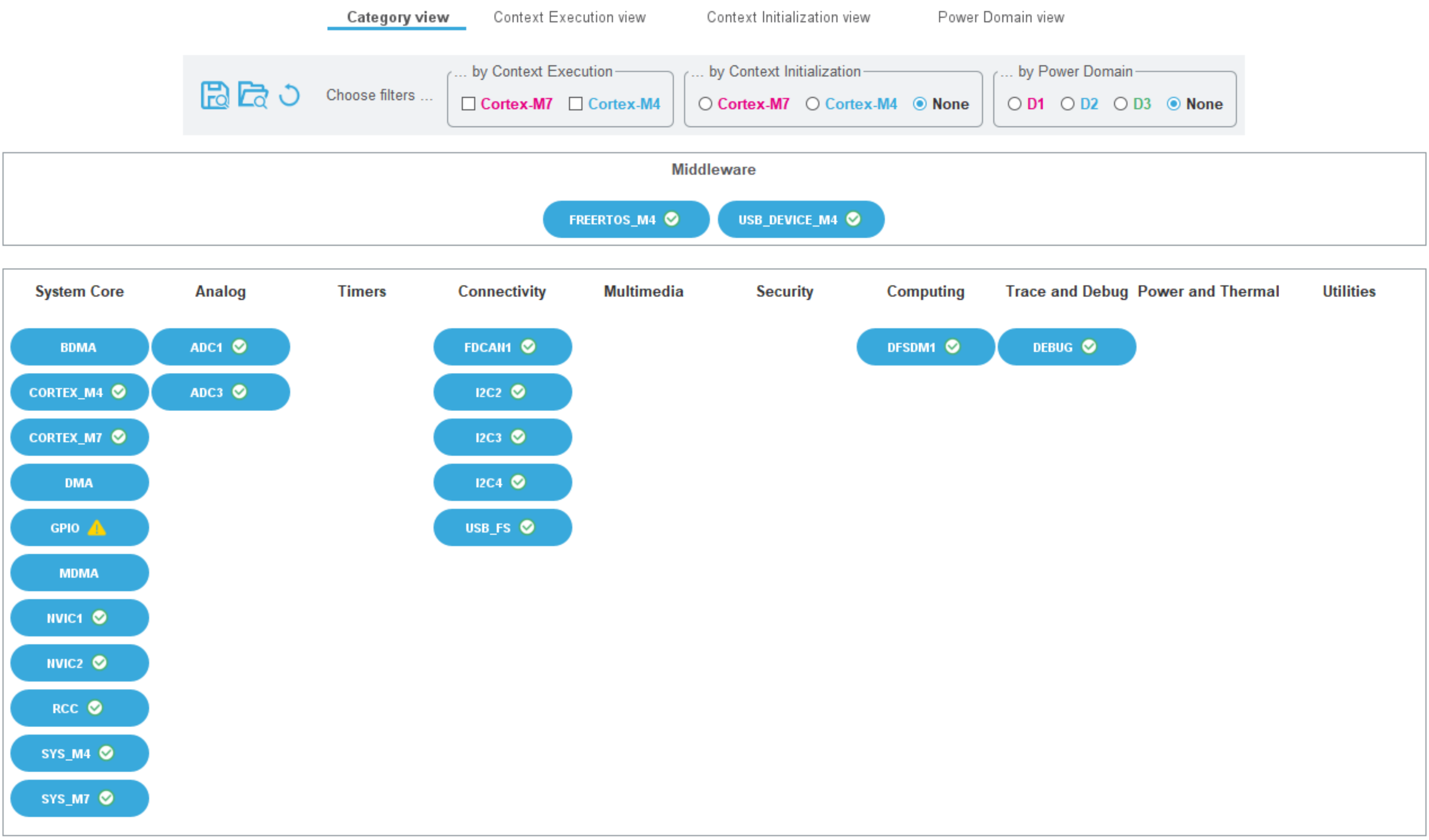
Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	true	true
USB On The Go FS global interrupt	false	true	true

*** User modified value**

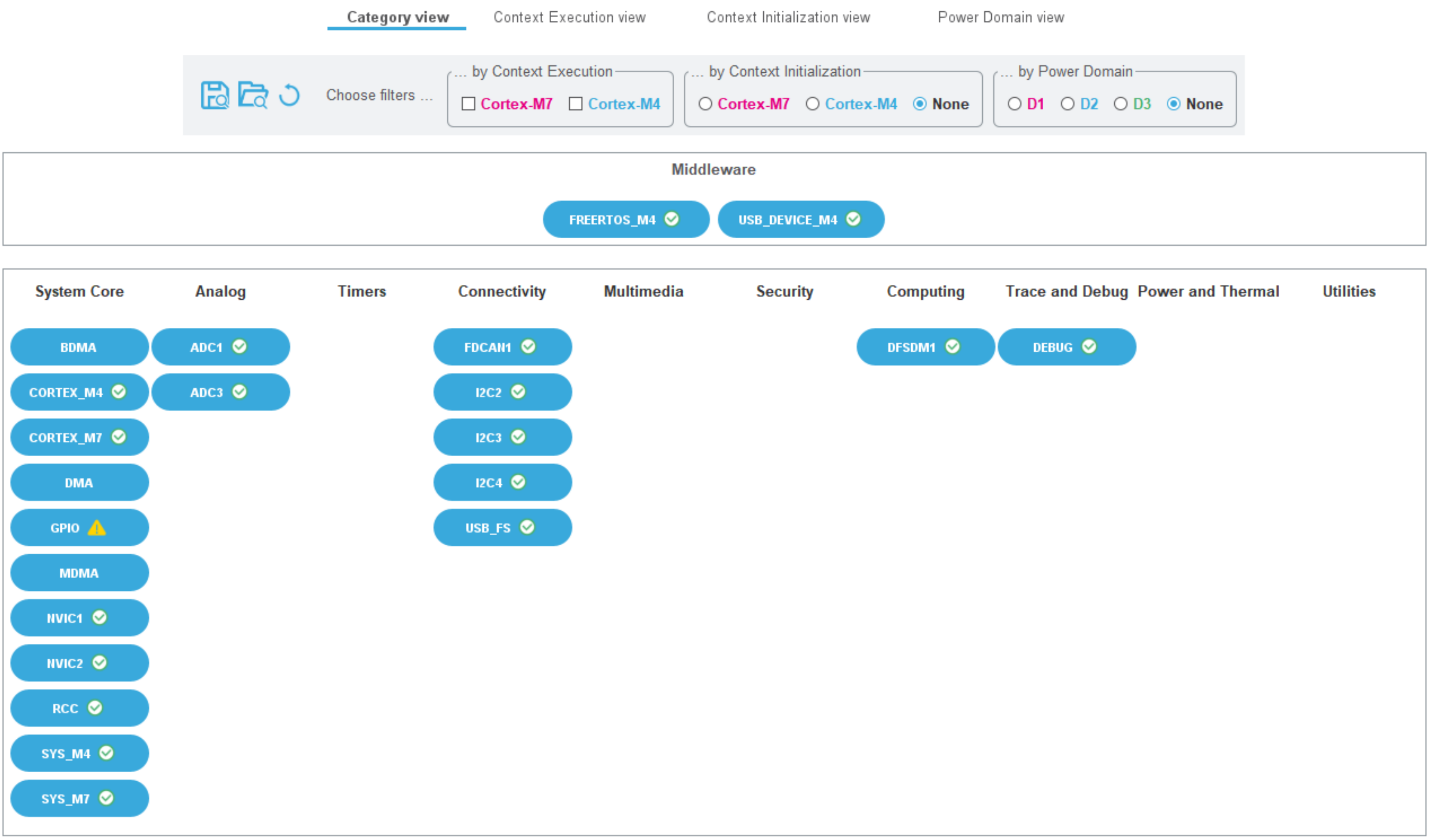
4. System Views

4.1. Category view

4.1.1. Current

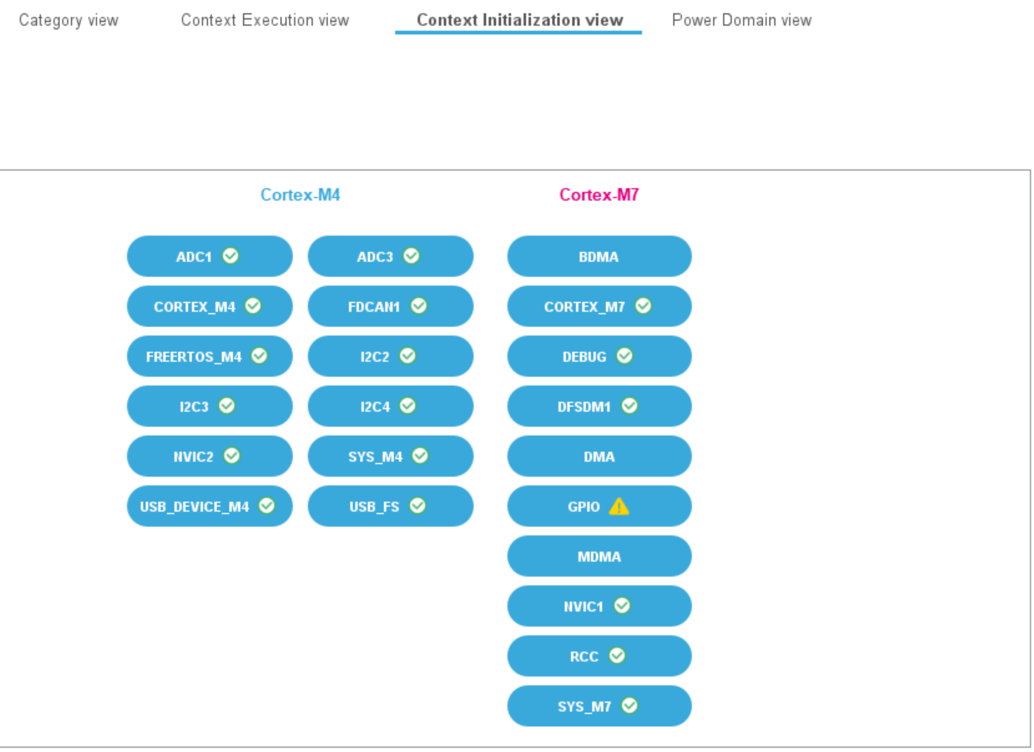


4.1.2. Without filters



4.2. Context Execution view

4.3. Context Initialization view



4.4. Power Domain view

5. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-

	guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelta-digital-interface-on-applicable-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5286-stm32h7x5x7-dualcore-microcontroller-debugging-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5354-getting-started-with-the-stm32h7-series-mcu-16bit-adc-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5557-stm32h745755-and-stm32h747757-lines-dualcore-architecture-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5617-stm32h745755-and-stm32h747757-lines-interprocessor-communications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4860-introduction-to-dsi-host-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-

	stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5342--how-to-use-error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4938-getting-started-with-stm32h74xig-and-stm32h75xig-mcu-hardware-development-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4992-introduction-to-secure-firmware-install-sfi-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4759-introduction-to-

	using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-with-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4861-introduction-to-lcdtft-display-controller-ltdc-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4943-how-to-use-chromart-accelerator-to-refresh-an-lcdtft-display-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5224-introduction-to-dmamux-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5507-how-to-use-crc-to-check-the-integrity-of-the-internal-flash-memory-on-stm32h7-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5337-guidelines-for-estimating-stm32h7-mcus-lifetime-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4996-how-to-use-the-jpeg-codec-peripheral-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5215-how-to-implement-advanced-power-management-on-stm32h747757-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-

	and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5020-introduction-to-digital-camera-interface-dcmi-for-stm32-mcus-stmicroelectronics.pdf
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